LITERATURE REVIEW · SMART DISTAP

Methods for image classification of plant stress

ML/DL techniques and image preprocessing for disease, stress, early-detection and time-point classification

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SCOPE

What this review looks for

Problems

- Disease classification
- Stress classification, any type: drought, nutrient, light, biotic
- Early detection and time-point classification: how soon, and how long since stress began

Techniques

- Image preprocessing and band selection
- Hand-made features with classical ML
- CNNs and transfer learning
- Hybrid CNN-LSTM, CNN-transformer and encoder-decoder models

Conditions of interest

- White-light and low-light imaging
- Controlled stress experiments, such as 4-hour exposures
- Data collected at several time points

Each paper is logged as: problem · ML/DL technique · reference · authors · year. Focus: RGB and three-band images; spectral work included for comparison.

OVERVIEW

Most studies follow the same five-step pipeline

1

Acquire

RGB or multi-band images, taken at one or several time points

2

Preprocess

Colour correction, background removal, resizing, augmentation

3

Represent

Hand-made colour and texture features, or features learned by a CNN

4

Model

Classical ML, CNN, CNN-LSTM for sequences, encoder-decoder

5

Output

Class (healthy, diseased, stressed), severity score or time point

The next slide explains the core methods; the tables after it group papers by problem.

FUNDAMENTALS

The core methods in plain terms

Method	What it does	Why it matters for us
Features + classical ML	People choose features (colour indices, texture); a classifier such as SVM or PLS-DA separates the classes	Simple baseline that works with little data
CNN	Learns features such as edges, colour changes and lesions directly from the image, then outputs a class	Takes any 3-channel image, so it fits our three-band images
Transfer learning	Starts from a CNN already trained on millions of everyday photos, then fine-tunes it on plant images	Our lab datasets are small
LSTM	Reads a sequence one step at a time and remembers what came before	Our images are taken at several time points
Hybrid CNN-LSTM	The CNN turns each time point's image into features; the LSTM links them across time	Main model; beat a CNN alone, 76.8% to 93% (Taghavi Namin et al., 2018)
Vision transformer (ViT)	Splits the image into patches and uses attention to weigh how each patch relates to the others	Can be combined with a CNN as a hybrid to explore
Encoder-decoder	The encoder compresses the input into a compact form; the decoder builds an output from it, such as a mask	To explore, e.g. for segmenting plants or learning features

MAP OF THE LITERATURE

Which techniques have been used for which problem

Problem	Hand-made features + classical ML	CNN and transfer learning	Detection and segmentation	Sequence models
Disease classification	None in this review	Mohanty 2016, Ferentinos 2018, Too 2019, Picon 2019, Borhani 2022	Fuentes 2017, Liu & Wang 2020, Lin 2019	None found
Abiotic stress	Naik 2017 (iron deficiency), Zhuang 2017 (water)	Ghosal 2018, An 2019, Chandel 2021, Azimi 2021a	None found	None found
Time-point / temporal	Sanchez 2020 (Raman, not images)	An 2019 (images every 2 h)	None found	CNN-LSTM: Taghavi Namin 2018, Azimi 2021b; Das Choudhury 2023
Preprocessing	Colour indices (Woebbecke 1995), colour calibration (Chopin 2018)	GAN augmentation (Arsenovic 2019)	Low-light enhancement (Ye 2024)	None found
Spectral comparison	SVM: Rumpf 2010; PLS-DA: Zovko 2019, Sanchez 2020	3D-CNN on hyperspectral (Nagasubramanian 2019)	None found	None found

All rows above "Spectral comparison" use RGB images unless marked. Detection papers use single images; sequence models use time-lapse images.

IMAGE PREPROCESSING

Preparing images before classification

Step	Method	Key finding	Reference
Plant/background segmentation	Chromatic indices r–g, g–b, ExG (2g–r–b), modified hue	ExG and hue separated plants in shaded and sunlit scenes	Woebbecke et al., 1995
Segmentation survey	Colour-index, threshold and learning-based methods	Overview of field segmentation options	Hamuda et al., 2016
Colour calibration	Colour chart in every image; quadratic RGB correction	Much lower day-to-day colour variation over 4 months	Chopin et al., 2018
Input variants	Colour vs grayscale vs leaf-segmented images	Colour input gave the best accuracy	Mohanty et al., 2016
Augmentation	Traditional transforms plus StyleGAN images	93.67% on a 79,265-image in-field set	Arsenovic et al., 2019
Low-light enhancement	Improved EnlightenGAN before YOLOv8 detection	mAP 95.26%, +2.47 pp over plain YOLOv8	Ye et al., 2024
Toolkit	PlantCV v2: normalisation, segmentation, ML modules	Open-source, high-throughput pipelines	Gehan et al., 2017

Challenges motivating these steps (complex backgrounds, uncontrolled capture): Barbedo, 2016

DISEASE CLASSIFICATION

CNNs with transfer learning dominate

Problem	ML/DL technique	Key result	Reference
38 classes, 14 crops (PlantVillage)	AlexNet, GoogLeNet; scratch vs transfer learning	99.35% on held-out test set	Mohanty et al., 2016
58 classes, 25 plants	AlexNet, GoogLeNet, Overfeat, VGG (best)	99.53%	Ferentinos, 2018
38 classes (PlantVillage)	Fine-tuned VGG16, Inception V4, ResNet, DenseNet-121	DenseNet best, 99.75%	Too et al., 2019
Cassava, field images	Transfer learning with a deep CNN	93% on unseen data	Ramcharan et al., 2017
Maize leaf blight, field	Patch CNNs make heat maps; final CNN classifies	96.7% on held-out images	DeChant et al., 2017
Wheat, mobile photos	Adapted deep residual network (ResNet)	Balanced accuracy 0.78 → 0.87	Picon et al., 2019
Multiple datasets	Vision Transformer vs CNN vs hybrid	Attention helps accuracy; hybrid restores speed	Borhani et al., 2022

Effect of dataset size and variety on transfer learning: Barbedo, 2018

STRESS CLASSIFICATION

Abiotic stress: from colour features to deep CNNs

Problem	ML/DL technique	Key result	Reference
Soybean, 8 biotic and abiotic stresses	Explainable CNN; top-K feature maps locate symptoms	94.13% on 16,207 images; also rates severity	Ghosal et al., 2018
Soybean iron-deficiency severity	10 classical classifiers on canopy colour features	Hierarchical classifier best, ~96%; phone app	Naik et al., 2017
Maize early water stress	Segmentation, colour and texture features, GBDT	Detects water stress early from digital images	Zhuang et al., 2017
Maize drought, 3 levels, field	Deep CNN vs GBDT; colour vs gray input	Identification 98.14%, classification 95.95%	An et al., 2019
Water stress: maize, okra, soybean	AlexNet, GoogLeNet, Inception-V3	GoogLeNet best: 98.3%, 97.5%, 94.1%	Chandel et al., 2021
Nitrogen-deficiency stress level	23-layer CNN vs ResNet18, NasNet-Large, classical ML	+8.25% average accuracy over classical ML	Azimi et al., 2021a

Review of deep learning for stress phenotyping: Singh et al., 2018

TIME-POINT CLASSIFICATION

Adding time improves accuracy

Problem	ML/DL technique	Key result	Reference
Arabidopsis genotype from time-lapse	CNN-LSTM on image sequences	CNN alone 76.8% → CNN-LSTM 93%	Taghavi Namin et al., 2018
Chickpea water stress over 5 months	CNN-LSTM on RGB shoot images	97.8–98.5%; ≥14% better than time-invariant models	Azimi et al., 2021b
Cotton drought onset	RGB time series + DTW; pixel CNN on hyperspectral	Predicted onset matches day stress appears	Das Choudhury et al., 2023
Maize drought, images every 2 h	Deep CNN on time-stamped field images	95.95% drought-level classification	An et al., 2019
Maize growth over time	Graph-based leaf/stem detection (not ML)	Leaf count and growth curves; public dataset	Das Choudhury et al., 2018
Rice N/P/K and salinity, by day	Hand-held Raman + PLS-DA (spectral)	N/P/K: 84.6% at day 2, 93.9% at day 4	Sanchez et al., 2020

No verified paper yet classifies hours after stress onset from RGB alone (see Gaps)

EARLY DETECTION

How much early-detection work exists, and of what type

9 hours

Earliest disease detection found: grey mould on tomato, multispectral sensor (Fahrentrapp et al., 2019)

3 to 4 days

Earliest drought and nutrient detection found: fluorescence at day 3 (Yao et al., 2018); hyperspectral at day 4 in choy sum (Teo et al., 2024)

16 papers by type

Type of work	Papers
Early disease detection	4
Early drought and salt stress	3
Early nutrient deficiency	3
Stress quantification (severity, days)	3
Shelf life and freshness	3

Imaging used: 10 spectral (hyperspectral, multispectral, fluorescence), 5 RGB only, 1 both.

No paper found detects abiotic stress within hours, and no RGB paper reports a pre-symptomatic detection time

EARLY DETECTION · DISEASE

Disease can be caught before symptoms appear

Problem	Imaging	ML/DL technique	Earliest detection	Reference
Tomato grey mould (detached leaves)	5-band multispectral	Regression, then classification	9 hours after infection; near-infrared and red-edge bands most useful	Fahrentrapp et al., 2019
Tobacco mosaic virus	Hyperspectral	Band selection + texture; neural nets, SVM	48 hours after infection, up to 95%; symptoms only at day 5	Zhu et al., 2017
Potato late and early blight	Leaf spectroscopy (not imaging)	Classification models	2 to 4 days before visible symptoms, over 80%	Gold et al., 2020
Tomato spotted wilt virus	Hyperspectral	GAN-based classifier	96.25% before visible symptoms	Wang et al., 2019

All four use spectral bands beyond normal colour; none uses RGB

EARLY DETECTION · ABIOTIC AND NUTRIENT STRESS

Drought, salt and nutrient stress: days, not hours

Problem	Imaging	ML/DL technique	Earliest detection	Reference
Arabidopsis drought	Chlorophyll fluorescence	Feature selection + SVM	93.3% at day 3; 99.1% at day 7	Yao et al., 2018
Okra salt stress, 13 genotypes	Hyperspectral	Deep-learning segmentation; regression	Imaged from day 2 of salt treatment	Feng et al., 2020
Choy sum N, P, K deficiency	Hyperspectral	Spectral-index feature mining (no classifier)	Day 4, before visible symptoms	Teo et al., 2024
Hemp nutrient deficiency	Hyperspectral	PLS-DA, SVM	Detected before visible symptoms	Sanaeifar et al., 2024
Pepper potassium deficiency	Hyperspectral, imaged hourly	XGBoost	80% at early growth stages	Weksler et al., 2021

Teo et al. (2024) studied the same crop as our first dataset, choy sum

QUANTIFICATION AND SHELF LIFE

Measuring how much stress, and how long produce lasts

Problem	Imaging	ML/DL technique	Key result	Reference
Soybean stress severity	RGB	Explainable CNN	Rates low, medium or high severity for 8 stresses	Ghosal et al., 2018
Cotton drought onset and spread	RGB time series + hyperspectral	Time-series matching; pixel-level CNN	Predicts the day stress starts and the daily % of plant stressed	Das Choudhury et al., 2023
Potato water stress level	Hyperspectral	RF, SVM, CNN, XGBoost and others	Estimates 4 stress levels, none to severe	Duarte-Carvajalino et al., 2021
Spinach freshness	RGB (smartphone)	Colour features; SVM, neural net	84% fresh vs not fresh, close to a human panel	Koyama et al., 2021
Persimmon shelf life	RGB	CNN with explanation maps	Over 80% for predicting over-softening	Suzuki et al., 2022
Mango storage time after bruising	Hyperspectral	CNN vs RF, PLS-DA, XGBoost	93.48% for storage time	Yao et al., 2024

RGB is used more here than for early detection, because severity and freshness are visible

DETECTION AND SEGMENTATION

Locating symptoms, not just labelling images

Problem	ML/DL technique	Key result	Reference
Tomato, 9 disease and pest classes	Faster R-CNN, R-FCN, SSD with VGG/ResNet backbones	R-FCN + ResNet-50 best, mAP 85.98%	Fuentes et al., 2017
Tomato, 10 diseases + 2 pests	YOLOv3 with image-pyramid multi-scale features	mAP 92.39%; 20 ms per image	Liu & Wang, 2020
Tea disease severity in low light	EnlightenGAN enhancement + improved YOLOv8	mAP 95.26%	Ye et al., 2024
Cucumber powdery mildew area	Pixel-level CNN semantic segmentation	Dice 83.45%; beat K-means, RF, GBDT	Lin et al., 2019
Leaf counting, age regression	Deep Plant Phenomics: pretrained CNN platform	State of the art on leaf counting	Ubbens & Stavness, 2017

Detection and segmentation also work as preprocessing: crop the leaf or lesion, then classify it

FOR COMPARISON: SPECTRAL ANALYSIS

Spectral work favours PLS-DA, SVM and 3D-CNNs

Problem	ML/DL technique	Key result	Reference
Sugar beet, early disease	SVM on hyperspectral reflectance	Early detection and classification	Rumpf et al., 2010
Maize drought, phenotyping platform	Close-range hyperspectral; SNV normalisation	Early stress detection in individual plants	Mohd Asaari et al., 2018
Soybean charcoal rot	Explainable 3D-CNN with saliency maps	95.73%; NIR bands most informative	Nagasubramanian et al., 2019
Grapevine drought	PLS-DA band selection, PLS-SVM	Drought vs irrigated >97%	Zovko et al., 2019
Rice nutrient and salinity stress	Hand-held Raman + PLS-DA	Salinity 96.9% at day 4	Sanchez et al., 2020
Nitrogen deficiency, 3 species	Portable Raman leaf-clip sensor	Early diagnosis; matched benchtop Raman	Gupta et al., 2020
Coleus: salinity, drought, chilling, light	Raman on stress pigments	Stress detected within 48 h	Altangerel et al., 2017

PLS-DA appears often; no PLSR-specific stress paper was verified in this pass

LIGHT CONDITIONS

Handling white-light and low-light images

Calibrate colour

A colour chart in every frame plus a quadratic RGB correction kept canopy colour consistent across months of changing light (Chopin et al., 2018).

Enhance dark images

EnlightenGAN low-light enhancement before YOLOv8 raised disease-severity mAP by 2.47 points (Ye et al., 2024).

Use robust colour indices

Normalised chromatic indices such as ExG separated plants under both shaded and sunlit conditions (Woebbecke et al., 1995).

Light as the stress

Raman detected light-saturation stress, among others, within 48 h (Altangerel et al., 2017). RGB evidence for light stress is thin.

Uncontrolled illumination is a key challenge for visible-range methods: Barbedo, 2016

GAPS

Where our experiments can add something new

Hour-level time points

No paper found detects drought, salt or nutrient stress within hours; the earliest is 3 to 4 days, and none with RGB.

White vs low light

Few studies compare the same model on white-light and low-light images of the same plants.

Which bands to use

Most reviewed papers use plain RGB. Whether three index-selected bands work better is open for our data.

Small lab datasets

Controlled experiments give few images, so transfer learning and augmentation are needed (Barbedo, 2018).

AGREED DIRECTION

Three-band inputs and hybrid CNN models

Input data

- A new index identifies three regions; one band is taken from each to build a three-band image
- Received: choy sum under high, low and white light, at 4 h and days 1 to 3
- Next: kailan, then the full dataset (including Arabidopsis) within 1 to 2 weeks

Model

- Hybrid models: CNN-LSTM as the main model, plus CNN with vision transformer
- Train and tune each crop separately, then on crops combined
- Also to explore: encoder-decoder architectures

NEXT: CODING

Coding plan

1. **Review the sample set.** Check the three-band images, labels and time points.
2. **Prepare data.** Organise by condition and time point; normalise; segment plants from background.
3. **Baselines.** Hand-crafted features with SVM or gradient boosting; a fine-tuned CNN (ResNet-50, DenseNet-121) on single images.
4. **Hybrid models.** CNN-LSTM on each plant's image sequence, plus a CNN-vision transformer; each crop first, then crops combined.
5. **Explore and evaluate.** Try an encoder-decoder variant; report accuracy per time point and condition, with saliency maps.

Altangerel to Das Choudhury

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Picon to Zovko

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Early detection and shelf life

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